Aalto University
School of Arts, Design
and Architecture

Check point no.3

Topic: Distributed Mixture-of-Experts (MoE) systems

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Topic Finalised

Routing-aware expert: Rebalancing for MoE training

MoE models are widely used for scalable LLM training

Performance strongly depends on:

- distributed execution
- expert placement
- communication efficiency

Real HPC clusters (CSC LUMI) provide an ideal platform for testing.

The topic combines:

- practical engineering
- performance analysis
- novel research contribution

Our Goal:
Improve efficiency of distributed MoE training beyond existing studies.

Phase 1 and Phase 2 Overview

Phase 1 – Benchmarking and Setup

Deploy MoE training on CSC/LUMI

Use existing frameworks:

- DeepSpeed **
- Megatron

Establish reproducible baseline results

Understand bottlenecks and limitations

Phase 2 – Innovation

Build routing-aware expert rebalancing

Integrate with training workflow

Evaluate improvement over baselines

Key idea:
First measure → then improve.

Choosing the Training Framework

Why DeepSpeed as Primary Framework

DeepSpeed provides support for:

- expert parallelism
- routing mechanisms
- profiling tools

DeepSpeed has:

- simpler setup
- active community and examples
- easier integration for modifications

Megatron will be considered only if needed

Personal and Team Objectives

After the Course We Aim To:

Be familiar with training MoE models

Benchmark DeepSpeed with open-source MoE models

Understand performance behavior of expert parallelism

Develop improved load-balancing strategies

Team Goals:

Produce reproducible benchmarks

Deliver a working rebalancing prototype

Benchmarking Plan

We will measure:

- Training throughput
- GPU utilization
- Memory usage
- Scaling efficiency
- Expert load imbalance
- Communication overhead

Specific Benchmarks

- Increase number of GPUs
  → Measure effect on throughput and memory
- Test different parallelism configurations
  → TP, DP., EP affects on metrics
- Analyze how configurations affect:
  → communication cost, resource usage

Work distribution

Person 1

Framework setup and deployment

Training process setup

Monitoring and data collection

Person 2

Benchmark design and metrics

Data analysis and visualization

Experiment planning

Both Students:

Design and implement rebalancing algorithms

Analyze experimental results

Write the final report

Timeline

Week 6-7

Environment setup on CSC/LUMI

Repository and shared documentation

Literature review

Week 8-9

Basic MoE training pipeline

Multi-GPU experiments

Initial benchmarks

Week 10-11

Rebalancing techniques implementation

Week 12-13

Experiments and evaluation

Week 14

Final report writing and deliverables packaged

Week 15

Demo and presentation

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Kiitos
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